

Homework for Chapter 7

- Study with the data from any source (reference books, websites, AI apps), and compare the transverse and longitudinal eigen-frequencies of optical phonons of a series of semiconductors of ZnO, ZnS, ZnSe, and ZnTe, what do you conclude?
- Compare the longitudinal-transverse splitting at $k = 0$ for Ge and the corresponding III–V, II–VI and I–VII compounds on the same row of the periodic table of the elements. What do you find and qualitatively explain your findings.

附录5 元素周期																		0	电子层	0 族 电子数
周期	I A	II A	III B	IV B	V B	VI B	VII B	8	9	10	I B	II B	III A	IV A	V A	VI A	VII A	18		
1	1 H 1.008	2 He 4.003																	2	
2	3 Li 6.941	4 Be 9.012	5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18											10	
3	11 Na 22.99	12 Mg 24.31	13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.06	17 Cl 35.45	18 Ar 39.95											18	
4	19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.87	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.41	31 Ga 69.72	32 Ge 72.64	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80	18	
5	37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.94	43 Tc [98]	44 Ru 101.1	45 Rh 102.9	46 Pd 106.4	47 Ag 107.9	48 Cd 112.4	49 In 114.8	50 Sn 118.7	51 Sb 121.8	52 Te 127.6	53 I 126.9	54 Xe 131.3	18	
6	55 Cs 132.9	56 Ba 137.3	57-71 La-Lu 镧系	72 Hf 178.5	73 Ta 180.9	74 W 183.8	75 Re 186.2	76 Os 190.2	77 Ir 192.2	78 Pt 195.1	79 Au 197.0	80 Hg 200.5	81 Tl 204.4	82 Pb 207.2	83 Bi 209.0	84 Po [209]	85 At [210]	86 Rn [222]	18	
7	87 Fr [223]	88 Ra [226]	89-103 Ac-Lr 锕系	104 Rf [261]	105 Db [262]	106 Sg [266]	107 Bh [264]	108 Hs [277]	109 Mt [268]	110 Uun [281]	111 Uuh [285]	112 Uub [285]	注：相对原子质量取自2001年国际原子量，并全部取4位有效数字。							
镧系	57 La 138.9	58 Ce 140.1	59 Pr 140.9	60 Nd 144.2	61 Pm [145]	62 Sm 150.4	63 Eu 152.0	64 Gd 157.3	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0					
锕系	89 Ac [227]	90 Th 232.0	91 Pa 231.0	92 U 238.0	93 Np [237]	94 Pu [244]	95 Am [243]	96 Cm [247]	97 Bk [247]	98 Cf [251]	99 Es [252]	100 Fm [257]	101 Md [258]	102 No [259]	103 Lr [262]					